DevOps Day 4 Task –

Kubernetes,Namespace:

Kubernetes (K8s)

Kubernetes is an open source container orchestration engine for automating deployment, scaling, and management of containerized applications. The open source project is hosted by the Cloud Native Computing Foundation (CNCF).
It provides a scalable and resilient framework for automating the deployment, scaling, and management of applications across clusters of servers.

A SMALL HISTORY OF K8S:

- In the early 2000s, Google started developing a system called Borg to manage their internal containerized applications.

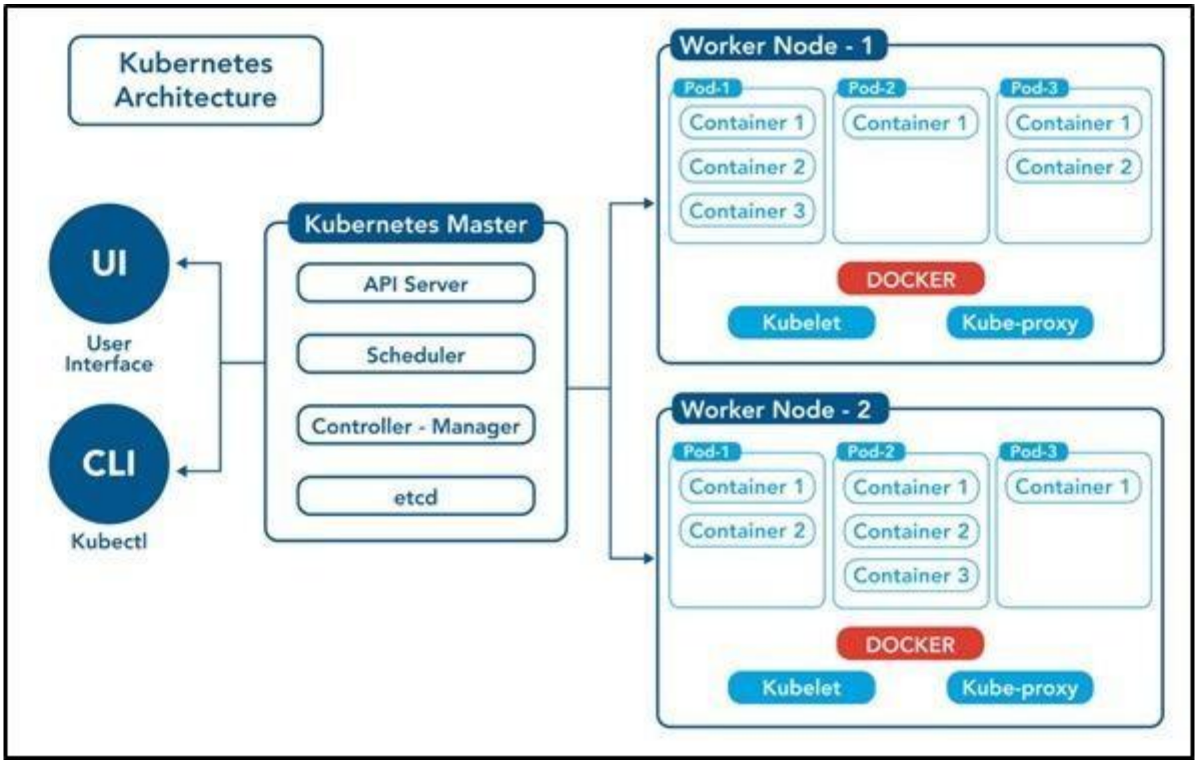
- Borg enabled Google to run applications at scale, providing features such as automatic scaling, service discovery, and fault tolerance.

- In 2014, Google open-sourced a version of Borg called Kubernetes.

- Kubernetes was donated to the Cloud Native Computing Foundation (CNCF), a neutral home for open-source cloud-native projects, in July 2015.

- Kubernetes 1.8 added significant enhancements for storage, security, and networking. Key features included the stable release of the stateful sets API, expanded support for volume plugins, and improvements in security policies.

- Check URL: https://kubernetes.io/releases/ for more release details.



Control Plane /Master Node

The control plane's components make global decisions about the cluster (for example, scheduling), as well as detecting and responding to cluster events (for example, starting up a new pod when a deployment's replicas field is unsatisfied).

Control plane components can be run on any machine in the cluster. Do not run user containers on this machine.

Node Components / Worker Nodes

Node components run on every node, maintaining running pods and providing the Kubernetes runtime environment.

1. Master Node: The master node is responsible for managing the cluster and coordinating the overall state of the system. It includes the following components:

a. API Server: The API server is the central control point for all interactions with the cluster. It exposes the Kubernetes API and handles requests from users and other components.

b. Scheduler: The scheduler is responsible for assigning workloads (pods) to individual worker nodes based on resource requirements, constraints, and other policies.

c. Controller Manager: The controller manager runs various controllers that monitor the cluster state and drive it towards the desired state. Examples include the replication controller, node controller, and service controller.

d. etcd: etcd is a distributed key-value store used by Kubernetes to store cluster state and configuration data.

1. Pod: The basic building block of Kubernetes. A pod represents a single instance of a running process within the cluster. It can encapsulate one or more containers that share the same network and storage resource

1. Create a pod using run command

$ kubectl run <pod-name> --image=<image-name> --port=<container-port>

$ kubectl run my-pod --image=nginx --port=80

2. View all the pods

(In default namespace)

$ kubectl get pods

(In All

namespace)

$ kubectl get pods -A

For a specific namespace

$ kubectl get pods -n kube-system

For a specific type

$ kubectl get pods <pod-name>

$ kubectl get pods <pod-name> -o wide

$ kubectl get pods <pod-name> -o yaml

$ kubectl get pods <pod-name> -o json

3. Describe a pod (View Pod details)

$ kubectl describe pod <pod-name>

$ kubectl describe pod my-pod

4. View Logs of a pod

$ kubectl logs <pod-name>

$ kubectl logs my-pod

5. Execute any command inside Pod (Inside Pod OS)

$ kubectl exec <pod-name> -- <command>

kubectl exec -it my-pod

[4:34 PM, 3/20/2025] +91 90928 13114: Namespace (short name = ns):

namespace is a virtual cluster or logical partition within a cluster that provides a way to organize and isolate resources. It allows multiple teams or projects to share the same physical cluster while maintaining resource separation and access control.

[4:34 PM, 3/20/2025] +91 90928 13114: # To create a namespace:

$ kubectl create namespace <namespace-name>

$ kubectl create ns my-bank

To switch to a specific namespace: (make this as default type)

$ kubectl config set-context --current --namespace=<namespace-name> #

To list all namespaces:

$ kubectl get namespaces

To get resources within a specific namespace:

$ kubectl get <resource-type> -n <namespace-name>

$ kubectl get deploy -n my-bank

$ kubectl get deploy --namespace my-bank

$ kubectl get all --namespace my-bank

To delete a namespace and all associated resources:

$ kubectl delete namespace <namespace-name>

$ kubectl delete ns my-bank

Deployment.yml

apiVersion: apps/v1

```yaml
kind: Deployment
metadata:
  name: my-deploy
  labels:
    name: my-deploy
spec:
  replicas: 1
  selector:
    matchLabels:
      apptype: web-backend
  strategy:
    type:
  RollingUpdate
  template:
    metadata:
      labels:
        apptype: web-backend
    spec:
      containers:
      - name: maven-web-app
        image: aswinprabusiva/webapp1:latest ports:
```

```
        - containerPort: 8000

---

apiVersion:
v1 kind:
Service
metadata:
  name: my-service
  labels:
    app: my-service
spec:
  type: NodePort
  ports:
    - port: 8000
      targetPort: 8080
      nodePort: 30007
```

```
pavis@pavish-kumar: ~
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Mar 20 03:30:02 UTC 2025

  System load:  0.46               Processes:             37
  Usage of /:   0.8% of 1006.85GB  Users logged in:       0
  Memory usage: 20%                IPv4 address for eth0: 172.19.33.247
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/pavis/.hushlogin file.
pavis@pavish-kumar:~$ minikube start
😄  minikube v1.35.0 on Ubuntu 24.04 (amd64)
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.46 ...
🔄  Restarting existing docker container for "minikube" ...
🐳  Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: default-storageclass, storage-provisioner
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
pavis@pavish-kumar:~$ |
```

Very humid Now | Search | ENG IN | 09:01 AM 20-03-2025

```
pavis@pavish-kumar: ~
pavis@pavish-kumar:~$ kubectl run my-pod --image=nginx --port=8083
pod/my-pod created
pavis@pavish-kumar:~$ kubectl get pods
NAME     READY   STATUS              RESTARTS   AGE
my-pod   0/1     ContainerCreating   0          37s
pavis@pavish-kumar:~$ kubectl get pods -A
NAMESPACE     NAME                               READY   STATUS              RESTARTS       AGE
default       my-pod                             0/1     ContainerCreating   0              58s
kube-system   coredns-668d6bf9bc-vcgkv           1/1     Running             6 (110s ago)   22h
kube-system   etcd-minikube                      1/1     Running             6 (110s ago)   22h
kube-system   kube-apiserver-minikube            1/1     Running             6 (110s ago)   22h
kube-system   kube-controller-manager-minikube   1/1     Running             6 (110s ago)   22h
kube-system   kube-proxy-982j6                   1/1     Running             6 (110s ago)   22h
kube-system   kube-scheduler-minikube            1/1     Running             6 (110s ago)   22h
kube-system   storage-provisioner                1/1     Running             10 (44s ago)   22h
pavis@pavish-kumar:~$ kubectl get pods -n kube-system
NAME                               READY   STATUS    RESTARTS      AGE
coredns-668d6bf9bc-vcgkv           1/1     Running   6 (2m4s ago)  22h
etcd-minikube                      1/1     Running   6 (2m4s ago)  22h
kube-apiserver-minikube            1/1     Running   6 (2m4s ago)  22h
kube-controller-manager-minikube   1/1     Running   6 (2m4s ago)  22h
kube-proxy-982j6                   1/1     Running   6 (2m4s ago)  22h
kube-scheduler-minikube            1/1     Running   6 (2m4s ago)  22h
storage-provisioner                1/1     Running   10 (58s ago)  22h
pavis@pavish-kumar:~$  kubectl get pods my-pod
NAME     READY   STATUS    RESTARTS   AGE
my-pod   1/1     Running   0          102s
pavis@pavish-kumar:~$ kubectl get pods my=pod -o wide
Error from server (NotFound): pods "my=pod" not found
pavis@pavish-kumar:~$ kubectl get pods my-pod -o wide
NAME     READY   STATUS    RESTARTS   AGE     IP            NODE       NOMINATED NODE   READINESS GATES
my-pod   1/1     Running   0          2m57s   10.244.0.10   minikube   <none>           <none>
pavis@pavish-kumar:~$ kubectl get pods my-pod -o yaml
```

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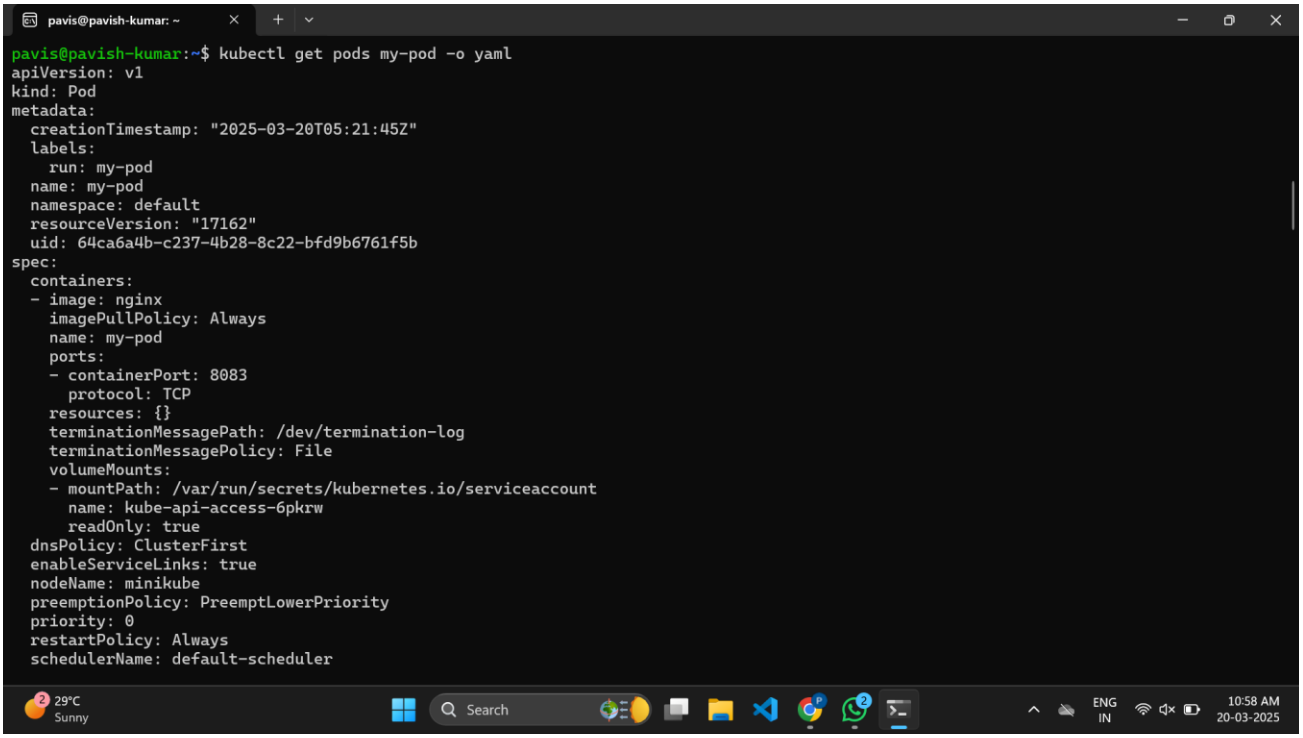
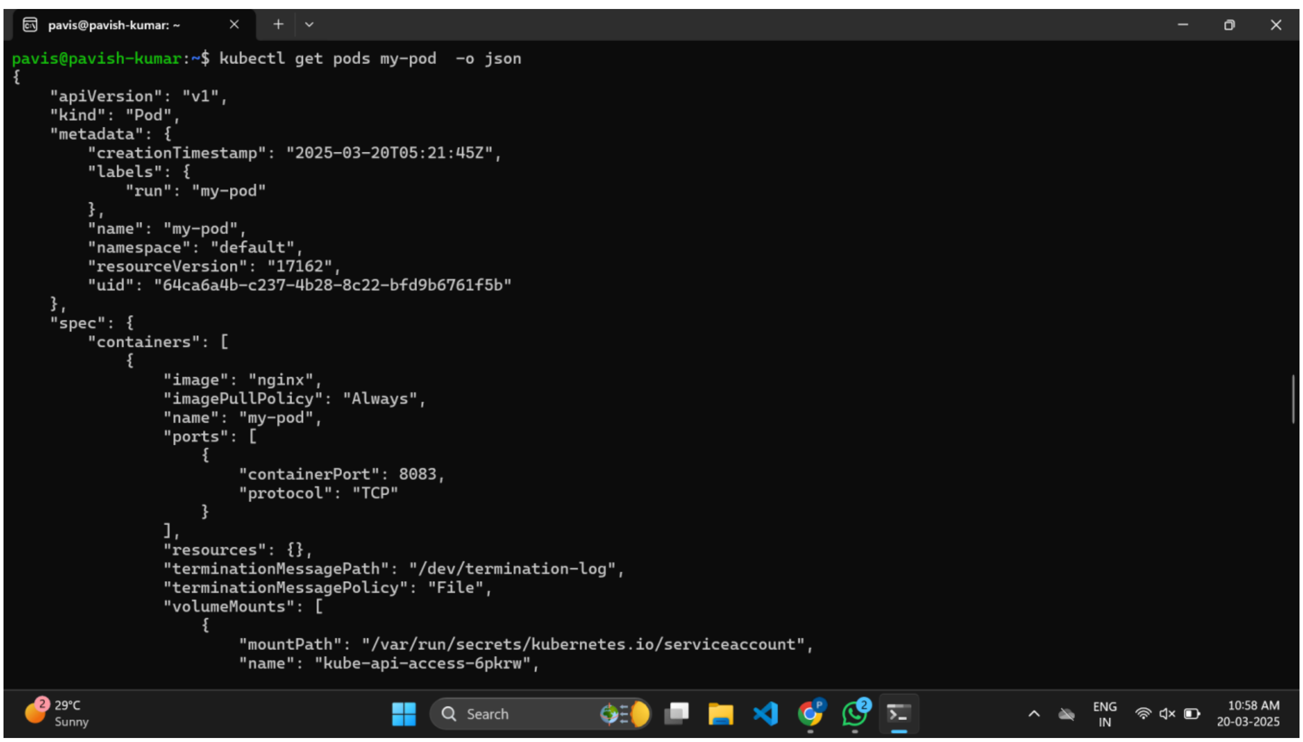



```
pavis@pavish-kumar:~$ kubectl get pods my-pod -o yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: "2025-03-20T05:21:45Z"
  labels:
    run: my-pod
  name: my-pod
  namespace: default
  resourceVersion: "17162"
  uid: 64ca6a4b-c237-4b28-8c22-bfd9b6761f5b
spec:
  containers:
  - image: nginx
    imagePullPolicy: Always
    name: my-pod
    ports:
    - containerPort: 8083
      protocol: TCP
    resources: {}
    terminationMessagePath: /dev/termination-log
    terminationMessagePolicy: File
    volumeMounts:
    - mountPath: /var/run/secrets/kubernetes.io/serviceaccount
      name: kube-api-access-6pkrw
      readOnly: true
  dnsPolicy: ClusterFirst
  enableServiceLinks: true
  nodeName: minikube
  preemptionPolicy: PreemptLowerPriority
  priority: 0
  restartPolicy: Always
  schedulerName: default-scheduler
```




```
pavis@pavish-kumar:~$ kubectl get pods my-pod  -o json
{
    "apiVersion": "v1",
    "kind": "Pod",
    "metadata": {
        "creationTimestamp": "2025-03-20T05:21:45Z",
        "labels": {
            "run": "my-pod"
        },
        "name": "my-pod",
        "namespace": "default",
        "resourceVersion": "17162",
        "uid": "64ca6a4b-c237-4b28-8c22-bfd9b6761f5b"
    },
    "spec": {
        "containers": [
            {
                "image": "nginx",
                "imagePullPolicy": "Always",
                "name": "my-pod",
                "ports": [
                    {
                        "containerPort": 8083,
                        "protocol": "TCP"
                    }
                ],
                "resources": {},
                "terminationMessagePath": "/dev/termination-log",
                "terminationMessagePolicy": "File",
                "volumeMounts": [
                    {
                        "mountPath": "/var/run/secrets/kubernetes.io/serviceaccount",
                        "name": "kube-api-access-6pkrw",
```

```
        },
        {
            "effect": "NoExecute",
            "key": "node.kubernetes.io/unreachable",
            "operator": "Exists",
            "tolerationSeconds": 300
        }
    ],
    "volumes": [
        {
            "name": "kube-api-access-6pkrw",
            "projected": {
                "defaultMode": 420,
                "sources": [
                    {
                        "serviceAccountToken": {
                            "expirationSeconds": 3607,
                            "path": "token"
                        }
                    },
                    {
                        "configMap": {
                            "items": [
                                {
                                    "key": "ca.crt",
                                    "path": "ca.crt"
                                }
                            ],
                            "name": "kube-root-ca.crt"
                        }
                    },
                    {
                        "downwardAPI": {
```


```
pavis@pavish-kumar:~$ kubectl describe pod my-pod
Name:             my-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.58.2
Start Time:       Thu, 20 Mar 2025 05:21:45 +0000
Labels:           run=my-pod
Annotations:      <none>
Status:           Running
IP:               10.244.0.10
IPs:
  IP:  10.244.0.10
Containers:
  my-pod:
    Container ID:   docker://89a852050554cad5cb51db3faa68bde2a06abb40e2d3be8e4675738198bb45db
    Image:          nginx
    Image ID:       docker-pullable://nginx@sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d3692baebc19
    Port:           8083/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Thu, 20 Mar 2025 05:23:22 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-6pkrw (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
```

```
pavis@pavish-kumar:~$ kubectl describe pod my-pod
Name:             my-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.58.2
Start Time:       Thu, 20 Mar 2025 05:21:45 +0000
Labels:           run=my-pod
Annotations:      <none>
Status:           Running
IP:               10.244.0.10
IPs:
  IP:  10.244.0.10
Containers:
  my-pod:
    Container ID:   docker://89a852050554cad5cb51db3faa68bde2a06abb40e2d3be8e4675738198bb45db
    Image:          nginx
    Image ID:       docker-pullable://nginx@sha256:124b44bfc9ccd1f3cedf4b592d4d1e8bddb78b51ec2ed5056c52d3692baebc19
    Port:           8083/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Thu, 20 Mar 2025 05:23:22 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-6pkrw (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
```

```
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-6pkrw (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-6pkrw:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  7m5s   default-scheduler  Successfully assigned default/my-pod to minikube
  Normal  Pulling    7m4s   kubelet            Pulling image "nginx"
  Normal  Pulled     5m29s  kubelet            Successfully pulled image "nginx" in 1m34.93s (1m34.93s including waiting). Image size
: 192004242 bytes.
  Normal  Created    5m28s  kubelet            Created container: my-pod
  Normal  Started    5m28s  kubelet            Started container my-pod
pavis@pavish-kumar:~$
```

pavis@pavish-kumar: ~

```
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  7m5s   default-scheduler  Successfully assigned default/my-pod to minikube
  Normal  Pulling    7m4s   kubelet            Pulling image "nginx"
  Normal  Pulled     5m29s  kubelet            Successfully pulled image "nginx" in 1m34.93s (1m34.93s including waiting). Image size
: 192004242 bytes.
  Normal  Created    5m28s  kubelet            Created container: my-pod
  Normal  Started    5m28s  kubelet            Started container my-pod
pavis@pavish-kumar:~$ kubectl logs my-pod
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/03/20 05:23:22 [notice] 1#1: using the "epoll" event method
2025/03/20 05:23:22 [notice] 1#1: nginx/1.27.4
2025/03/20 05:23:22 [notice] 1#1: built by gcc 12.2.0 (Debian 12.2.0-14)
2025/03/20 05:23:22 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/03/20 05:23:22 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/03/20 05:23:22 [notice] 1#1: start worker processes
2025/03/20 05:23:22 [notice] 1#1: start worker process 29
2025/03/20 05:23:22 [notice] 1#1: start worker process 30
2025/03/20 05:23:22 [notice] 1#1: start worker process 31
2025/03/20 05:23:22 [notice] 1#1: start worker process 32
2025/03/20 05:23:22 [notice] 1#1: start worker process 33
2025/03/20 05:23:22 [notice] 1#1: start worker process 34
2025/03/20 05:23:22 [notice] 1#1: start worker process 35
2025/03/20 05:23:22 [notice] 1#1: start worker process 36
pavis@pavish-kumar:~$
```

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pavis@pavish-kumar: ~

```
2025/03/20 05:23:22 [notice] 1#1: start worker process 32
2025/03/20 05:23:22 [notice] 1#1: start worker process 33
2025/03/20 05:23:22 [notice] 1#1: start worker process 34
2025/03/20 05:23:22 [notice] 1#1: start worker process 35
2025/03/20 05:23:22 [notice] 1#1: start worker process 36
pavis@pavish-kumar:~$ kubectl exec my-pod -- run
error: exec [POD] [COMMAND] is not supported anymore. Use exec [POD] -- [COMMAND] instead
See 'kubectl exec -h' for help and examples
pavis@pavish-kumar:~$ kubectl exec my-pod
error: you must specify at least one command for the container
pavis@pavish-kumar:~$ kubectl exec my-pod -- ls
bin
boot
dev
docker-entrypoint.d
docker-entrypoint.sh
etc
home
lib
lib64
media
mnt
opt
proc
root
run
sbin
srv
sys
tmp
usr
var
pavis@pavish-kumar:~$
```

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```
pavis@pavish-kumar:~$ sudo nano my-pod.yaml
pavis@pavish-kumar:~$ kubectl apply -f my-pod.yaml
pod/my-app created
pavis@pavish-kumar:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   23h   v1.32.0
pavis@pavish-kumar:~$ kubectl get pods
NAME     READY   STATUS              RESTARTS   AGE
my-app   0/1     ContainerCreating   0          18s
my-pod   1/1     Running             0          45m
pavis@pavish-kumar:~$
```

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